

Multiple Choice (10 marks)

1. A rod measures 1.00 m in its rest system. How fast must an observer move parallel to the rod to measure its length to be 0.80 m?
 - (a) 0.50 c
 - (b) 0.60 c
 - (c) 0.70 c
 - (d) 0.80 c
2. For which of the following thermodynamic processes is the increase in the internal energy of an ideal gas equal to the heat added to the gas?
 - (a) Constant temperature
 - (b) Constant volume
 - (c) Constant pressure
 - (d) Adiabatic
3. In the diamond structure of elemental carbon, the nearest neighbors of each C atom lie at the corners of a
 - (a) square
 - (b) hexagon
 - (c) cube
 - (d) tetrahedron
4. The rest mass of a particle with total energy 5.0 GeV and momentum 4.9 GeV/ c is approximately
 - (a) 0.1 GeV/ c^2
 - (b) 0.2 GeV/ c^2
 - (c) 0.5 GeV/ c^2
 - (d) 1.0 GeV/ c^2
5. The energy required to remove both electrons from the helium atom in its ground state is 79.0 eV. How much energy is required to ionize helium (i.e., to remove one electron)?
 - (a) 24.6 eV
 - (b) 39.5 eV
 - (c) 51.8 eV
 - (d) 54.4 eV

True / False (10 marks)

Answer True or False

6. True or False: Doubling the voltage across a resistor results in a new rate of energy dissipation of 2 W.
7. True or False: The electric displacement current through a surface is proportional to the rate of change of the electric flux through that surface.
8. True or False: Bosons have symmetric wave functions and obey the Pauli exclusion principle, allowing multiple bosons to occupy the same state.
9. True or False: The speed of light in a nonmagnetic dielectric material with a dielectric constant of 4.0 is 1.5×10^8 m/s.
10. True or False: A helium-neon laser is the best type of laser for spectroscopy over a range of visible wavelengths due to its stability.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the behavior of unpolarized light as it passes through two ideal linear polarizers with their transmission axes at a 45-degree angle. What is the intensity of the transmitted light relative to the incident intensity, and explain the underlying principles that lead to this result.
12. Discuss the role of dopants in germanium semiconductors, specifically analyzing which ions are suitable for creating n-type semiconductors and why some are not.

End of Paper